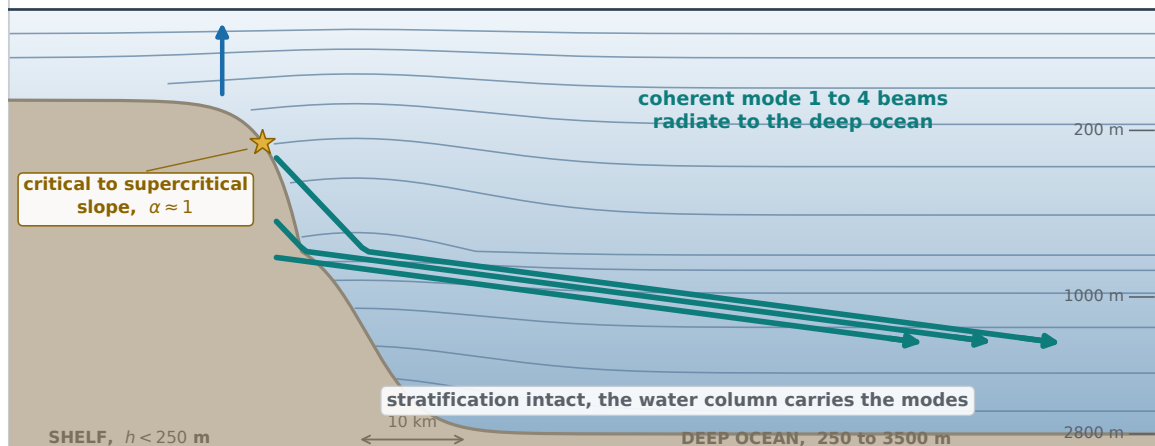


Unforced

tidal forcing only

barotropic M_2 tide, spring

phase-locked tidal upwelling
at the shelf break, $S < 0$



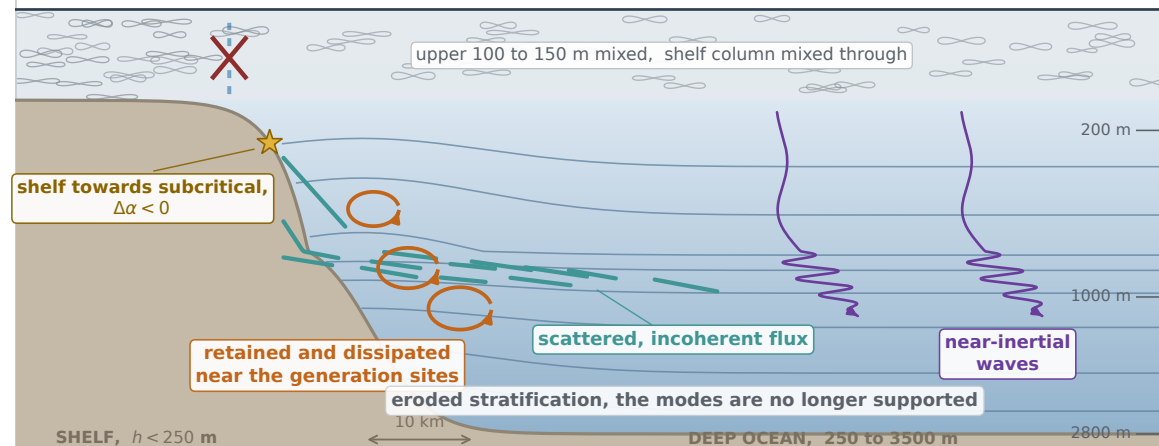
Wind-forced

Hurricane Catarina, 24 to 30 March 2004

westward track, landfall 28 March

tidal upwelling weakened,
reversed at the storm peak

wind stress



M_2 conversion, domain total

295 → 220 MW

Deep-ocean export, after the passage

63% → 15%

Retained near generation, after the passage

37% → 85%

Tidal-to-supertidal transfer, after the wind relaxed

6.4× the unforced level